Real-Time Pollution Detection for IIoT: A Hybrid Reinforcement Learning Offloading Framework for Multi-Tier Satellite Networks

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***Abstract*—Continuous and real-time pollution detection in industrial environments within the Industrial Internet of Things (IIoT) is crucial for maintaining ecological balance and ensuring the safe operation of industrial facilities. However, traditional pollution detection methods suffer from limitations in coverage and data timeliness. In recent years, the rapid development of satellite remote sensing technology has provided new opportunities to address these challenges. To this end, we propose an edge computing framework integrating satellite remote sensing with IIoT. This approach deploys deep learning models for pollution detection directly on satellites, enabling real-time on-orbit analysis and processing of captured images and downlinking only key detection results (e.g., pollution location, type, and extent), thereby significantly enhancing response speed. We first outline the architecture and processing workflow of this edge computing system, followed by the development of a hybrid Advantage Actor-Critic (A2C) and Decision Transformer (DT) reinforcement learning algorithm. Experimental results demonstrate that the proposed method dynamically optimizes the trade-off between processing accuracy and latency in pollution detection tasks under strict delay constraints, thus meeting the urgent demands for high timeliness and wide coverage in IIoT scenarios.**

***Index Terms*—Industrial Internet of Things (IIoT), Orbital Edge Computing (OEC), Pollution Detection, Satellite Remote Sensing, Reinforcement Learning**

I. Introduction

As humanity rapidly advances toward the Industry 4.0 era, the proliferation of the Industrial Internet of Things (IIoT) has not only enhanced the intelligence and automation of production processes but also exacerbated industrial pollution challenges [1]. Issues such as gas leaks, water contamination, and industrial waste discharges pose ongoing threats to ecological balance and industrial safety. Consequently, achieving real-time and efficient pollution detection in IIoT has garnered increasing attention.

Currently, pollution detection in IIoT mainly relies on ground-based sensor networks deployed in industrial sites (e.g., factories, pipelines, and storage tanks) to collect real-time data on gas concentrations, water quality parameters, and related metrics. This vast amount of monitoring data is aggregated at regional IIoT gateways and forwarded to ground-based cloud computing centers for processing [2]. Additionally, unmanned aerial vehicles (UAVs), IoT devices, and low-altitude sensing platforms serve as supplementary tools to extend detection coverage and perform inspections in critical areas. In recent years, advances in remote sensing technology have enabled the use of satellite imagery and deep learning methods (e.g., object detection and semantic segmentation) as effective tools for large-scale, non-contact pollution monitoring [3]. However, these methods exhibit limitations in real-time performance and coverage. Ground-based networks struggle to cover remote areas with weak infrastructure (e.g., offshore regions, mountains, and deserts), while the traditional paradigm of on-board acquisition, satellite-to-ground transmission, and ground-based processing incurs significant delays in data transmission and analysis, hindering timely responses to sudden pollution events.

To address these limitations, deploying edge computing capabilities on satellites—enabling a paradigm of on-board acquisition, on-orbit computing, and result downlink—has emerged as an effective solution [4] [5]. By integrating lightweight deep learning models directly on satellites, captured remote sensing imagery can be analyzed promptly, with only key pollution incident details (e.g., location coordinates, type, and extent) downlinked. This approach not only reduces bandwidth demands on satellite-to-ground links but also shifts processing to the satellite side, creating an on-board intelligent layer that integrates seamlessly with ground-based IIoT, thus enabling real-time pollution detection in remote, isolated, and emergency scenarios.

We propose a satellite-based edge computing framework for IIoT pollution detection. Leveraging low Earth orbit (LEO) and medium Earth orbit (MEO) satellites along with inter-satellite links, an on-orbit computing network is established. Upon receiving detection requests from the IIoT gateways, remote sensing satellites capture imagery, which is processed on-board via slice and task offloading decisions at edge nodes, returning only the final pollution results.

II. Related Work

With rapid population growth and industrial development, monitoring and automatically detecting industrial pollution, especially marine pollution and air pollution, has become one

of the most urgent issues in the Industrial Internet of Things (IIoT). Real-time and effective detection of industrial pollution poses a significant challenge. Facing complex industrial environments, traditional detection methods such as manual investigation and visual inspection are inadequate in terms of scalability, real-time applicability, and accuracy. The rapid advances in deep learning for automatic object detection and segmentation have provided promising solutions for the growing demand for industrial pollution detection. These models are typically deployed on the ground, receiving remote sensing imagery acquired and downlinked by remote sensing satellites or images captured by unmanned aerial vehicle (UAV) aerial networks as input, and performing pollution detection by learning pixel-level feature representations. Sawant et al. [6] compare the performance of different YOLO variants for rapid detection of marine pollution. Biju and Ramasamy [7] use YOLOv8 and YOLOv11 for object detection and multiple CNNs for classification, achieving high accuracy in classifying three different types of microplastic pollution in water bodies. Kurah et al. [8] perform oil spill pollution detection in dynamic port environments based on DeepLabV3.

Directly deploying lightweight object detection models or semantic segmentation models on satellites to construct Orbital Edge Computing (OEC) networks is more conducive to real-time response in industrial detection task processing, addressing the limitation that ground-based processing cannot detect remote areas. Reinforcement learning has achieved significant progress in the field of service offloading, which can greatly improve system performance, efficiency, and resource utilization [9]. Chen et al. [10] propose a deep reinforcement learning-based algorithm for dynamic resource management in edge computing for IIoT. Doan et al. [11] propose a novel method based on Q-learning that utilizes the concept of adjacent state sequences. Hammoud et al. [12] adopt an edge computing architecture and perform oil spill detection based on U-Net processing of UAV-captured data.

III. System Model

As illustrated in Fig. 1, we consider a satellite–edge-enabled IIoT system comprising a space segment and a ground segment that cooperate to perform real-time pollution detection. In the space segment, LEO remote-sensing satellites acquire high-resolution imagery, MEO orchestration satellites slice images and orchestrate subtask offloading, and LEO edge-computing satellites perform on-orbit inference. The ground segment comprises IIoT gateways (which issue requests and receive results), ground stations (which provide uplink/downlink access), and ground cloud computing centers (which offer virtually unlimited computing resources and serve as fallback nodes when latency constraints are relaxed). End-to-end, task requests are uplinked to space, imagery is captured and forwarded to MEO for slicing and scheduling, selected nodes execute inference, and compact results are returned to the gateway.

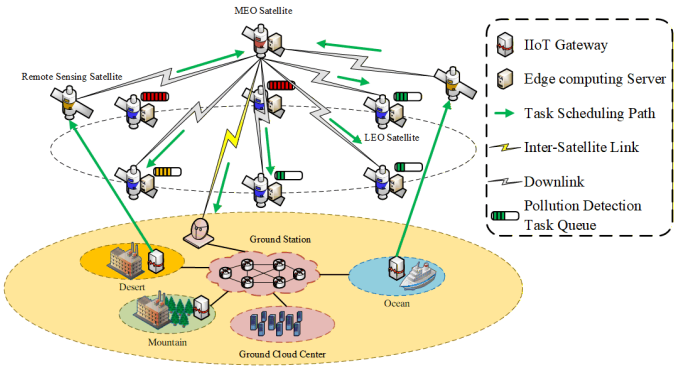


Fig. 1: Framework of the proposed satellite-edge-enabled IIoT pollution detection system.

A. Communication Model

The communication performance of the proposed system relies on two primary link types: inter-satellite links (ISLs) and satellite-to-ground links (SGLs).

1) Inter-Satellite Links (ISLs): For analytical tractability and to focus on the high-level decision-making in the subsequent reinforcement learning framework, we adopt a simplified model for the ISL transmission rate:

$$C_{\text{ISL}} = C_0 + \mathcal{N}(0, \sigma^2), \tag{1}$$

where C_0 is the nominal rate and the Gaussian term captures random fluctuations due to channel dynamics.

2) Satellite-to-Ground Links (SGLs): Satellite-to-ground links are primarily employed for task offloading to ground cloud computing centers when required.

The slant range d between a satellite at altitude h and a ground station with elevation angle θ is given by:

$$d = \sqrt{h^2 + 2hR_e(1 - \sin\theta)}, \tag{2}$$

where R_e is the Earth's radius.

The path loss $PL(d)$ of the satellite-to-ground downlink is a key indicator of signal attenuation, computed as specified by the 3GPP NTN specification [13]:

$$PL(d) = 32.45 + 20\log_{10}(f_0) + 20\log_{10}(d) + F_s + L_g + L_s, \tag{3}$$

where $PL(d)$ denotes the total path loss (in dB), f_0 is the carrier center frequency, F_s represents shadow fading, L_g is atmospheric absorption loss, and L_s is ionospheric scintillation loss.

The received power is:

$$P_R = P_T + G_t(\theta, \phi) + G_R - PL(d), \tag{4}$$

where P_T is the satellite transmit power, and $G_T(\theta, \phi)$ and G_R are the transmit and receive antenna gains, respectively.

The carrier-to-noise ratio (CNR) is:

$$\text{CNR} = \frac{P_R}{N_0}, \tag{5}$$

where $N_0 = k_B \cdot T_{\text{sys}} \cdot B$ is the noise power, with k_B the Boltzmann constant, T_{sys} the system noise temperature, and B the bandwidth.

B. Mobility Model

Simulations are conducted in discrete time slots. Within each time slot, the network topology, inter-satellite link states, and relative satellite positions are assumed static, whereas they change significantly across slots due to orbital motion.

The orbit of each satellite s is described by the classical Keplerian elements: $\Psi_s = \{a_s, e_s, i_s, \nu_s, \omega_s, \Omega_s\}$, which allow precise computation of the satellite's position in the Earth-centered inertial (ECI) frame at any time t using standard orbital propagation methods.

LEO satellites have short orbital periods (typically 90–120 min), resulting in brief ground-station visibility windows and rapidly changing inter-satellite link topologies [14]. Using LEO nodes as central schedulers would lead to frequent handovers, unstable collaboration, and potential failure to return results when satellites leave coverage.

To ensure scheduling stability, we designate MEO satellites as the core decision nodes. Relative to LEO constellations and ground infrastructure, MEO satellites exhibit much slower topological variation, providing longer-term resource visibility and more stable inter-satellite links. This design significantly enhances the reliability of task slicing, offloading, and result delivery in the orbital edge-computing framework.

C. Computation Model

1) Task and Data Model: The proposed system processes a sequence of pollution detection tasks initiated by IIoT gateways and forwarded to remote sensing satellites. Each task corresponds to a raw remote sensing image with dimensions $J \times W \times H$, where J is the number of spectral bands, and W and H denote the width and height in pixels, respectively. When transmitted via inter-satellite links, each pixel is quantized to u bits, resulting in a data volume of $J \times W \times H \times u$ bits per raw image.

Upon receiving the raw image m at the MEO decision-making satellite, a segmentation strategy $\pi_m^{\text{seg}} = (k, o)$ is selected, where k is the number of sub-blocks and o is the overlap (in pixels) between adjacent sub-blocks. Each sub-block has size $\bar{W} \times \bar{H}$. The number of sub-blocks k is determined by:

$$k = \left\lceil \frac{W - o}{\bar{W} - o} \right\rceil \times \left\lceil \frac{H - o}{\bar{H} - o} \right\rceil, \tag{6}$$

which accounts for the reduced stride due to overlap.

The MEO satellite also makes an offloading decision for each sub-task. This decision is encapsulated in the vector $\pi_m^{\text{ass}} = (n_m^1, n_m^2, \ldots, n_m^{k_m})$, where each $n_m^i \in \{0, 1, \ldots, S\}$ indicates the computing node selected for the i-th sub-task.

2) Computation Load: All computing nodes employ the DeepLabV3+ model for semantic segmentation. Let F_m denote the number of floating-point operations (FLOPs) required to process one pixel during inference. The computational load of a single sub-block is $F_m \cdot u \cdot J \cdot \bar{W} \cdot \bar{H}$. Each node maintains a task queue containing the computational loads (in FLOPs) of pending subtasks.

3) Latency Model: For subtask i of task m, the total processing latency T_i depends on the assigned computing node. The latency expressions for the three cases are as follows:

(a) When processed on the MEO satellite:

$$T_{i,\text{MEO}} = T_{\text{queue}} + \frac{F_m u J \bar{W} \bar{H}}{G_{\text{MEO}}}, \tag{7}$$

(b) When offloaded to a LEO satellite:

$$T_{i,\text{LEO}} = \frac{u J \bar{W} \bar{H}}{C_{\text{ISL}}} + T_{\text{queue}} + \frac{F_m u J \bar{W} \bar{H}}{G_{\text{LEO}}}, \tag{8}$$

(c) When offloaded to the ground cloud center:

$$T_{i,\text{grd}} = \frac{u J \bar{W} \bar{H}}{C_{\text{SGL}}} + T_{\text{queue}} + \frac{F_m u J \bar{W} \bar{H}}{G_{\text{grd}}}. \tag{9}$$

In all cases, the latency comprises up to three components: transmission delay (present only for offloaded subtasks), queuing delay at the target node, and computation delay.

The total latency of task m is determined by the slowest subtask, as subtasks are processed in parallel:

$$T_m = \max_i \{T_i\}. \tag{10}$$

4) Detection Accuracy Model: To quantify detection effectiveness, we use mean Intersection over Union (mIoU) from the DeepLabV3+ semantic segmentation output. Each task is associated with images from the public Potsdam dataset. We pre-train the DeepLabV3+ model on the training split until convergence.

To establish the relationship between segmentation strategy and accuracy, we apply each candidate strategy $\pi^{\text{seg}} = (k, o)$ to the fixed test split of the Potsdam dataset, perform inference, and record the resulting mIoU as $\eta(\pi^{\text{seg}})$. The detection accuracy A_m for task m is thus:

$$A_m = \eta(\pi^{\text{seg}}). \tag{11}$$

This offline mapping enables the trade-off between accuracy and latency in the subsequent optimization.

D. Problem Formulation

At each decision epoch, the MEO orchestration satellite receives a raw remote sensing image corresponding to pollution detection task m. It must jointly determine:

(a) a tile-slicing strategy $\pi_m^{\text{seg}} = (k_m, o_m)$, where k_m is the number of tiles and o_m is the overlap in pixels,

(b) an offloading decision vector $\pi_m^{\text{ass}} = (n_m^1, n_m^2, \ldots\ldots, n_m^{k_m})$, where n_m^i indicates the computing node assigned to tile i.

Let $\pi_m = \left(\pi_m^{\text{seg}}, \pi_m^{\text{ass}}\right)$ denote the complete joint decision for task m. The resulting end-to-end latency and detection accuracy are denoted as $T_m(\pi_m) = \max\{T_i\}$ and $A_m(\pi_m) = \eta(\pi_m^{\text{seg}})$, respectively.

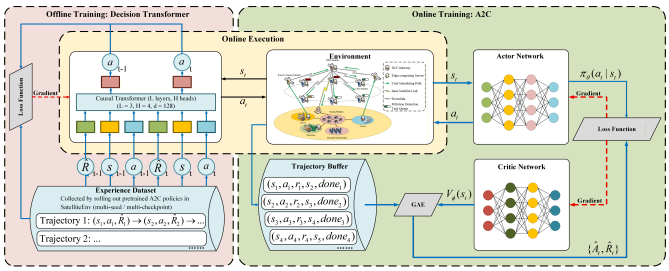


Fig. 2: Framework of the Proposed Hybrid A2C-DT Network for Satellite Edge Computing

Our objective is to maximize long-term detection accuracy while satisfying per-task latency constraints in the highly dynamic satellite environment. The global optimization problem is formulated as:

$$\max_{\{\pi_m\}} \quad \lim_{M\to\infty} \frac{1}{M} \sum_{m=1}^{M} A_m(\pi_m) \tag{12}$$

$$\text{s.t.} \quad T_m(\pi_m) \le T_{\max}, \quad \forall m \tag{C1}$$

$$n_m^i \in \{0, 1, \ldots, S\} \quad \forall i = 1, \ldots, k_m \tag{C2}$$

$$k_m \in \mathbb{Z}^+, \quad o_m \in \{0, 1, \ldots, \min(\bar{W}, \bar{H}) - 1\} \tag{C3}$$

where (C1) enforces the stringent end-to-end latency requirement for industrial pollution detection, (C2)–(C3) define the discrete feasible sets for offloading and slicing decisions,

The above problem is a constrained sequential decision-making problem with mixed-integer actions, time-varying network states, and nonlinear latency–accuracy coupling. Traditional optimization methods cannot efficiently solve it due to the curse of dimensionality and lack of future state information. Therefore, in the next section, we propose a deep reinforcement learning approach to learn near-optimal joint slicing and offloading policies online.

IV. Hybrid A2C-DT Joint Optimization Algorithm

This section details our hybrid training paradigm that integrates online exploration with offline distillation to tackle the dynamic joint slicing and offloading problem in satellite edge computing. We first formulate the problem as a Markov Decision Process (MDP) to capture its sequential nature. An online Advantage Actor-Critic (A2C) agent is then employed to learn a policy while generating a rich dataset of experience trajectories within a high-fidelity simulator. Finally, a Decision Transformer (DT) distills a more robust and temporally coordinated policy from this dataset through supervised sequence modeling. This paradigm combines the sample-efficient exploration of online reinforcement learning with the stability and long-horizon reasoning capability of offline training, confining all exploratory risk to simulation and yielding a policy ready for safe, low-risk deployment in real orbital environments.

A. Markov Decision Process (MDP) Formulation

We model the joint task-slicing and offloading problem as a Markov Decision Process (MDP), with the MEO orchestration satellite acting as the learning agent. The MDP is defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma)$, where $\gamma \in (0, 1)$ is the discount factor.

(a) State Space $\mathcal{S}$: At time slot t, the agent observes the global state

$$s_t = \{\mathcal{G}_t, \mathcal{Q}_t, \mathcal{L}_t, \mathcal{T}_t\}, \tag{13}$$

where $\mathcal{G}_t$ represents the current satellite constellation topology and ground station visibility, $\mathcal{Q}_t$ contains the queue backlog (in FLOPs) of all computing nodes, $\mathcal{L}_t$ represents the instantaneous ISL and SGL rates, and $\mathcal{T}_t$ includes the attributes of the incoming task (image dimensions $W \times H$, number of bands J, and latency threshold $T_m^{\text{threshold}}$).

(b) Action Space $\mathcal{A}$: The joint action is

$$a_t = (\pi_t^{\text{seg}}, \pi_t^{\text{ass}}) = (k_t, o_t, n_{t,1}, \ldots, n_{t,k_t}), \tag{14}$$

where (k_t, o_t) is the selected slicing strategy and $n_{t,i}$ denotes the assigned node for tile i.

(c) Reward Function r: To strictly enforce the latency constraint while maximizing detection accuracy, the immediate reward is defined as

$$r_t = \begin{cases} \eta(\pi_t^{\text{seg}}), & \text{if } T_t(\pi_t) \le T_{\text{threshold}}, \\ 0, & \text{otherwise.} \end{cases} \tag{15}$$

This hard-constraint reward encourages the agent to learn feasible policies that maximize mIoU only when the end-to-end latency requirement is satisfied.

The objective is to find a policy $\pi(a|s)$ that maximizes the expected long-term discounted return $\mathbb{E}[\sum_{t=0}^{\infty} \gamma^t r_t]$.

B. Online Exploration: Policy Learning and Experience Collection via A2C

We employ the Advantage Actor-Critic (A2C) algorithm to solve the formulated MDP and to generate the necessary experience dataset for subsequent offline optimization by the DT, as illustrated in Fig. 2.

The proposed A2C decision network consists of two components:

(a) Actor Network: Takes the global state s_t as input and outputs a probability distribution over the joint action space. For slicing parameters (k_t, o_t), we use independent categorical distributions; for offloading decisions of the k_t tiles, we adopt a variable-length softmax output with masking to handle different k_t.

(b) Critic Network: Estimates the state-value function $V(s_t; \theta_v)$ to reduce variance in policy gradient updates.

At each time slot t, the agent selects action $a_t \sim \pi(\cdot|s_t; \theta_\pi)$, executes it in the environment, and receives reward r_t and next state s_{t+1}. The advantage function is computed as

$$\hat{A}_t = r_t + \gamma V(s_{t+1}; \theta_v) - V(s_t; \theta_v). \tag{16}$$

The Critic is updated by minimizing the temporal-difference(TD) loss:

$$\mathcal{L}_v = \mathbb{E}\left[\hat{A}_t^2\right]. \tag{17}$$

The Actor is updated using the policy gradient with entropy regularization:

$$\mathcal{L}_\pi = -\mathbb{E}\left[\hat{A}_t \log \pi(a_t|s_t;\theta_\pi) + \beta H(\pi(\cdot|s_t))\right], \tag{18}$$

where the entropy term encourages exploration. This process enables stable learning of effective decision policies in the dynamic satellite environment.

C. Offline Distillation: Sequence-Based Policy Coordination via DT

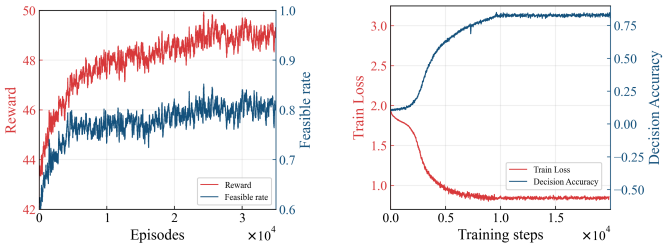
As shown in Fig. 2, the target problem is reformulated as a sequence decision problem using the collected experience trajectory dataset from the previous stage. Each trajectory is represented as a sequence of states, actions, and returns:

$$\tau = (s_1, a_1, R_1, s_2, a_2, R_2, \dots), \tag{19}$$

where $R_t = \sum_{k=t}^{\infty} \gamma^{k-t} r_k$ denotes the discounted return-to-go from time step t. The input to the DT consists of the embedded representation of this sequence along with a target return $\hat{R}$ as a conditioning signal to guide the generation of actions that achieve the desired return. Specifically, the model leverages self-attention to capture long-term dependencies in the sequence and predicts the action that best achieves the desired return. This allows the model to account for long-term decision consequences and improves cross-slot coordination.

The training dataset is constructed from multiple A2C checkpoints and random seeds to ensure diversity. The DT is trained by minimizing the cross-entropy loss between its predicted actions and the expert actions from the trajectories, performing supervised offline learning. This process distills a coordination-aware policy that exploits long-term patterns beyond the step-wise horizon of A2C.

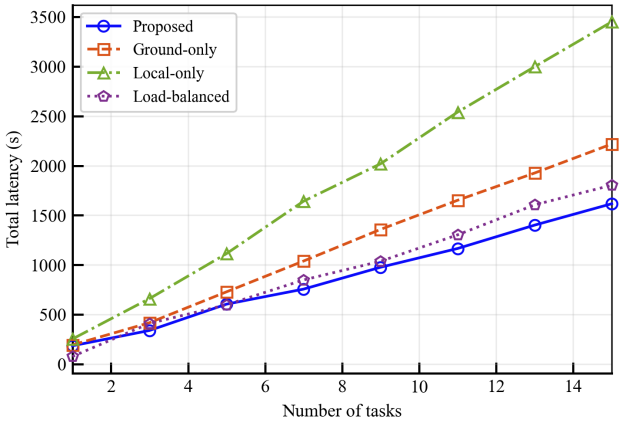
V. Experimental Results and Analysis



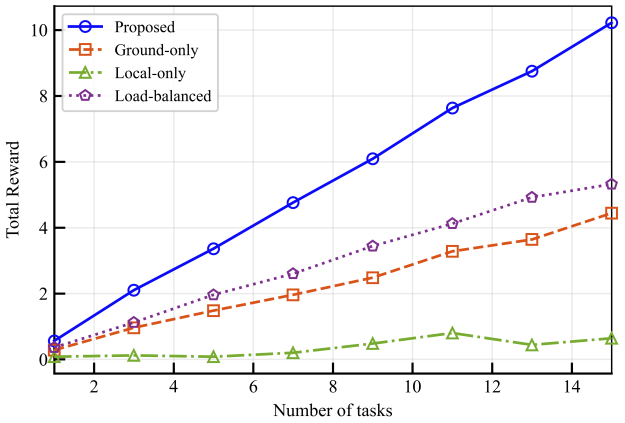
(a) A2C (b) Decision Transformer

Fig. 3: Training performance of the Proposed Hybrid A2C-DT Network. (a) A2C online exploration: reward and feasible rate across episodes. (b) DT offline optimization: train loss and decision accuracy across training steps.

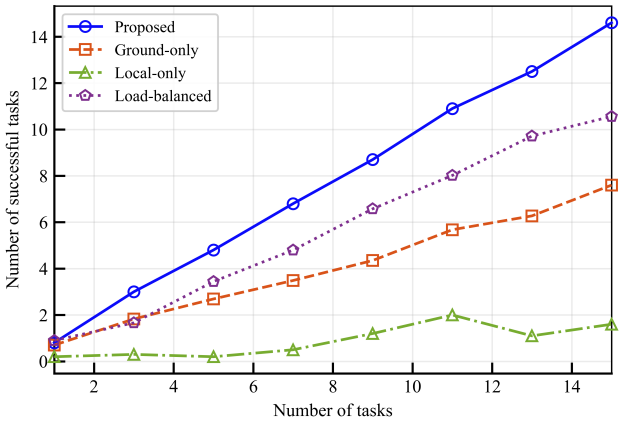
The experiments are conducted using a custom discrete-time simulator implemented in Python with PyTorch. The



(a) Total latency versus task scale for different algorithms



(b) Cumulative reward versus task scale for different algorithms



(c) Number of successful tasks versus task scale for different algorithms. Each marker represents the mean over multiple evaluation episodes.

Fig. 4: Performance comparison of different algorithms: (a) total latency, (b) cumulative reward, and (c) successful tasks versus different task scales. Algorithms: Proposed (Hybrid A2C-DT joint slicing/offloading), Ground-only (raw image downlink), Local-only (MEO-only processing), Load-balanced (round-robin distribution).

TABLE I: Key Simulation Parameters

Parameter	Value
Communication	
ISL frequency	26.5 GHz
Downlink frequency	12.0 GHz
ISLs per satellite	6
Additional space loss	5.5 dB
EIRP density	41.41 dBW/MHz
Shadow fading std. dev.	2 dB
Satellite Constellation	
Remote sensing satellites	Gaofen-like (20)
LEO edge computing satellites	Starlink-like (33)
MEO orchestration satellites	Galileo-like (24)
Computation	
On-board capacity (LEO/MEO)	150 GFLOPS
Ground cloud capacity	14 TFLOPS
Task and Model	
Raw image size	6000×6000 pixels
Bits per pixel	24
Spectral bands	3
Inference model	DeepLabV3+

simulator integrates realistic orbital dynamics, inter-satellite and satellite-to-ground link budget calculations, dynamic task queue management, and DeepLabV3+ inference modeling. Key simulation parameters are summarized in Table I.

The training progress is shown in Fig. 3. The A2C agent (Fig. 3(a)) successfully learns a policy that maximizes reward while maintaining a high task feasibility rate (consistently above 80%), demonstrating its ability to satisfy strict latency constraints. Subsequently, the DT (Fig. 3(b)) efficiently distills this knowledge, as evidenced by its rapidly decreasing loss and high decision accuracy, confirming effective sequence modeling.

As shown in Fig. 4, the proposed method consistently attains the highest accumulated reward, maintains the lowest average end-to-end latency per task, and completes the most deadline-satisfying tasks across all task scales, with its advantage growing under heavier loads due to adaptive slicing and offloading. In contrast, baseline strategies are limited by fixed constraints: Ground-only by downlink capacity, Local-only by on-orbit compute, and Load-balanced by ignoring node heterogeneity. These limitations lead to progressively worse performance at scale, widening the gap with our approach.

VI. Conclusion

In this paper, we propose a hybrid A2C-DT reinforcement learning paradigm for IIoT pollution detection, addressing coverage and timeliness issues in industrial pollution detection through orbital edge computing, on-orbit processing, task slicing, and offloading. By first modeling the problem as an MDP and employing A2C for online exploration and experience collection, and then leveraging DT for offline distillation of coordination-aware policies, our method synergistically combines sample efficiency with long-horizon decision-making. Simulations confirm that the proposed method outperforms baselines in accuracy and latency, ensuring reliable task completion. Future work involves advanced models, hardware deployment, and multi-modal data, enabling scalable remote monitoring.

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